

Nimbus Nine — SD1 Training Pilot

Workbook 1

Import, Inspect, and Spot Patterns

Difficulty	2 / 10 (warm-up)
Estimated time	30 – 45 minutes

Files for this workbook

Start file: 02_workbook_data/workbook1_start.xlsx

Complete file: 02_workbook_data/workbook1_complete.xlsx

Source files required:

- 01_source_data/source_primary.xlsx
- 01_source_data/source_secondary.csv
- 01_source_data/source_lookup.pdf

FROM THE FLIGHT DECK

Engineer Pip Quasar says:

“Importing data is half the job. Take it slow. We are not racing anyone.”

YOU ARE DEVELOPING

Skills: S1 (collect data), S3 (organise data), S6 (use software tools)

Knowledge: K7 (data sources), K8 (data formats), K10 (data quality)

Behaviours: B2 (works methodically), B4 (responds positively to feedback)

The mission

MISSION BRIEF

Captain Vega has a question:

“Which cargo routes are running smoothly, and which routes need a little help?”

Before anyone can answer, we need to bring our three source files into Excel cleanly.

Your job in this workbook is to import them using Power Query, take a calm look around, and write down what you notice.

What you will learn

- How to open Power Query through Excel's Data tab.
- How to import an Excel file, a CSV file, and a PDF file.
- How to load each query as a Connection Only.
- How to spot small data oddities without panicking.

Why this matters

WHY THIS MATTERS

Power Query is Excel's tool for importing, cleaning and reshaping data.

Building the Power Query habit early means cleaner, faster work for the rest of your career.

Captain Vega trusts a learner who stops to look at the data before changing it.

Mission 1 — Open the start workbook

We will open the workbook that has been prepared for you. It already has a Welcome tab and an Inspection_Notes tab.

1. Open Excel.
2. Click File, then Open, then Browse.
3. Navigate to the folder 02_workbook_data.
4. Double-click the file workbook1_start.xlsx.
5. Confirm two tabs are visible at the bottom: Welcome and Inspection_Notes.

PAUSE AND CHECK

Can you see the tabs Welcome and Inspection_Notes?

If yes, you are good to keep going.

If no, close Excel and try again. The file lives in 02_workbook_data/workbook1_start.xlsx.

Mission 2 — Import the primary Excel file

We are going to bring source_primary.xlsx into Power Query.

Step by step

1. On the Excel ribbon, click the Data tab.
2. Click Get Data.
3. Hover on From File.
4. Click From Excel Workbook.
5. In the file picker, navigate to 01_source_data/source_primary.xlsx.
6. Double-click source_primary.xlsx. The Navigator window will appear.
7. In Navigator, click the sheet named Deliveries on the left.
8. Click Transform Data (not Load).
9. Power Query Editor will open in a new window.

Tidy the query name

10. On the right-hand side, in the Query Settings panel, find the Name box.
11. Change the name to Deliveries_Raw.
12. Press Enter.

Quick checks before loading

13. Read the column names along the top. There should be eight columns.
14. Read the row count at the bottom of the window. It should be 52 rows.
15. Click Home, then Close & Load, then Close & Load To.
16. Choose Only Create Connection.
17. Click OK.

PAUSE AND CHECK

On the right of Excel, the Queries & Connections panel should now show one query named Deliveries_Raw.

Connection only is exactly what we want for now.

If you do not see the panel, click Data, then Queries & Connections.

COMMON TRAP

Clicking Load instead of Transform Data is a very common slip.

It will pour the rows into a sheet straight away. That is not what we want.

If it happens, click Undo (Ctrl + Z) and start the Mission again.

This catches lots of people. Mistakes are normal.

Mission 3 — Import the CSV file

Now we will import source_secondary.csv. CSV stands for comma-separated values. It is just plain text with commas between fields.

1. Click Data on the ribbon.
2. Click Get Data.
3. Hover on From File.
4. Click From Text/CSV.
5. Navigate to 01_source_data/source_secondary.csv.
6. Double-click source_secondary.csv.
7. A preview window appears. Check the delimiter is set to Comma.
8. Confirm there are 5 columns and 20 data rows visible in the preview.
9. Click Transform Data.
10. In Query Settings on the right, rename the query to Couriers_Raw.
11. If the first row contains your column names, click Home, then Use First Row as Headers.
12. Click Home, then Close & Load, then Close & Load To.
13. Choose Only Create Connection. Click OK.

PAUSE AND CHECK

Open Data > Queries & Connections.

You should now see two queries: Deliveries_Raw and Couriers_Raw.

Both should say Connection only. Lovely.

Mission 4 — Import the PDF lookup

Now we will pull the routes table out of a PDF. Yes, Power Query can do that.

1. Click Data on the ribbon.
2. Click Get Data.
3. Hover on From File.
4. Click From PDF.
5. Navigate to 01_source_data/source_lookup.pdf.
6. Double-click source_lookup.pdf.
7. The Navigator window appears. You will see one or more items listed.
8. Click each item in turn until you see a 5-column table that looks like the routes (route_code, route_name, sector, base_distance_au, difficulty). Select that one.
9. Click Transform Data.
10. In Query Settings on the right, rename the query to Routes_Raw.
11. Click Home, then Close & Load, then Close & Load To.
12. Choose Only Create Connection. Click OK.

PAUSE AND CHECK

Open Data > Queries & Connections.

You should now see three queries: Deliveries_Raw, Couriers_Raw, Routes_Raw.

All three should say Connection only.

COMMON TRAP

PDF imports can list several tables. Sometimes the right one is called Table001 or Page001.

If yours looks different, try the next one. That is a clue, not a failure.

Mission 5 — Reopen your queries

We are going to make sure you can find a query again. This habit will save you in every workbook.

1. Click the Data tab.
2. Click Queries & Connections (if the panel is not already open).
3. Find Deliveries_Raw in the list on the right.
4. Double-click the name Deliveries_Raw.
5. Power Query Editor should open with the Deliveries_Raw query loaded.
6. Close the editor by clicking Home, then Close & Load.

NICE WORK

You just opened a query straight from the Queries & Connections panel.
From now on, this is how you will return to your work between workbooks.

Mission 6 — Spot simple patterns

Now for the fun bit. We are going to peek at each query and write down what we notice. We are not changing anything yet — just observing.

Look at Deliveries_Raw

1. In Queries & Connections, double-click Deliveries_Raw.
2. Scroll down through the data preview.
3. Look at the status column. Do you see any odd casing such as 'delivered' or 'DELIVERED'?
4. Look at the cargo_type column. Do any values seem to have spaces around them?
5. Look at the parcel_weight_kg column. Are there any blank cells?

Look at Couriers_Raw

6. Select Couriers_Raw.
8. Look at the courier_name column. Any leading spaces hiding there?
9. Look at the ship_class column. Any odd casing?

Look at Routes_Raw

10. Select Routes_Raw.
11. How many routes are listed? (Should be 8.)
12. Compare the route codes (R01, R02, etc.) to anything you saw in Deliveries_Raw.
13. Is there a route in Deliveries_Raw that does not appear here? (There should be one — that is fine for now.)
14. Close the editor with Home, then Close & Load.

Write your notes

15. Click the Inspection_Notes tab in workbook1_start.xlsx.
16. Fill in a few rows with what you spotted.
17. There is no perfect answer — calm observation is what matters.

TRY THIS

Tiny challenge:

Find one observation that is not a problem — something the data is doing well.

It might be that the email column looks consistent, or fees look sensible.

Confidence grows when we notice the good things too.

What you can now do

- Open Power Query through Data > Get Data.
- Import an Excel workbook and choose a sheet in Navigator.
- Import a CSV file and check the delimiter.
- Import a PDF and select the correct table.
- Load a query as Connection Only.
- Reopen a query from Queries & Connections.
- Spot small patterns in raw data without panicking.

NICE WORK

That was Workbook 1.

Three queries imported, three quiet observations made.

If you want to compare against the worked version, open
02_workbook_data/workbook1_complete.xlsx.

Captain Vega is already proud of you.